Introduction to agents

Weekend 6 · More Agentic AI

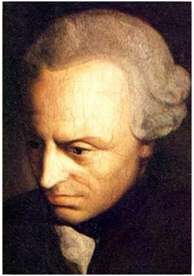
Carlos Cotrini

Section 1

Where we are

1

All the interest of my reason



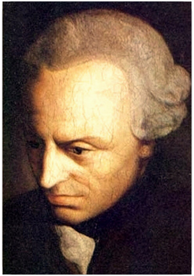
Immanuel Kant (c. 1790), painter unknown

Alles Interesse meiner Vernunft (das spekulative sowohl, als das praktische) vereinigt sich in folgenden drei Fragen:

1.
2.
3.

Kritik der reinen Vernunft, A805 / B833

All the interest of my reason



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1. Was kann ich wissen?
2.
3.

What can I know?

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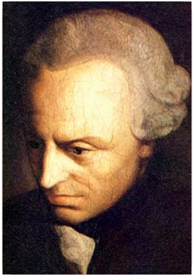

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1. Was kann ich wissen?
2. Was soll ich tun?
3.

What can I know? What ought I to do?

Kritik der reinen Vernunft, A805 / B833

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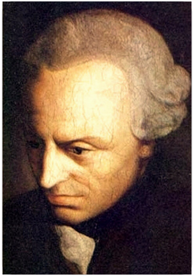

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2. Was soll ich tun?
3. Was darf ich hoffen?

What can I know? What ought I to do? What may I hope?

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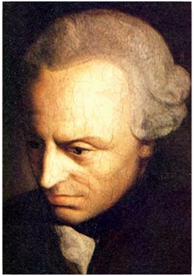
All the interest of my reason



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What can I know? What ought I to do? What may I hope?

Kritik der reinen Vernunft, A805 / B833

Kant asked these of himself. We will ask them of the machine.

The same three questions, asked of the machine

Was kann *es* wissen? → Weekend 5: RAG

Was soll *es* tun? → Weekend 6: Agentic AI

Was darf *es* hoffen? → Weekend 7: Large Scale AI

Weekend 5 was what it can know. This weekend is what it should do.

Two days

Friday

- Agents, RL, and MCTS
- LATS
- Agentic AI in software engineering
- Intergalactic football tournament

Two days

Friday

- Agents, RL, and MCTS
- LATS
- Agentic AI in software engineering
- Intergalactic football tournament

Saturday

- Agent Q, and the limits of agentic AI
- Evolutionary coding agent
- Project

Two days

Friday

- Agents, RL, and MCTS ← **we are here**
- LATS
- Agentic AI in software engineering
- Intergalactic football tournament

This block is two half hours. The first is this lecture; the second is Monte Carlo tree search.

Saturday

- Agent Q, and the limits of agentic AI
- Evolutionary coding agent
- Project

From a table to an agent

the model

Reasoning model	trained by RL to reason before answering
Large language model	a transformer compresses any prefix
Language model	predict the next word

From a table to an agent

prompting	Tree-of-Thought	branch, score, keep the best
	Chain-of-Thought	write the reasoning first
the model	Reasoning model	trained by RL to reason before answering
	Large language model	a transformer compresses any prefix
	Language model	predict the next word

From a table to an agent

agents	LATS	all of them, inside a search	at 10:00
	Monte Carlo Tree Search	look ahead before moving	**at 08:30**
	ReAct	think, act, observe	
prompting	Tree-of-Thought	branch, score, keep the best	this lecture
	Chain-of-Thought	write the reasoning first	
the model	Reasoning model	trained by RL to reason before answering	
	Large language model	a transformer compresses any prefix	
	Language model	predict the next word	

Section 2

Language models

2

language model → LLM → reasoning model → tree of thought → agents

A language model is a table

Next word

coffee	tea	is	hot	cold	⟨end⟩

Columns: the vocabulary, every word the model can say

A language model is a table

Prefix so far	Next word: coffee	tea	is	hot	cold	⟨end⟩
nothing yet						
coffee						
tea						
coffee is						
tea is						
coffee is hot						
⋮						

Rows: the prefix, everything said so far

A language model is a table

Prefix so far	Next word: coffee	tea	is	hot	cold	⟨end⟩
nothing yet	0.6	0.4	0	0	0	0
coffee	0	0	1.0	0	0	0
tea	0	0	1.0	0	0	0
coffee is	0	0	0	0.8	0.2	0
tea is	0	0	0	0.5	0.5	0
coffee is hot	0	0	0	0	0	1.0
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Entries: how likely each word is to come next

A language model is a table

Prefix so far	Next word: coffee	tea	is	hot	cold	⟨end⟩
nothing yet	0.6	0.4	0	0	0	0
coffee	0	0	1.0	0	0	0
tea	0	0	1.0	0	0	0
coffee is	0	0	0	0.8	0.2	0
tea is	0	0	0	0.5	0.5	0
coffee is hot	0	0	0	0	0	1.0
⋮	⋮	⋮	⋮	⋮	⋮	⋮

$$\Pr(\text{hot} \mid \text{coffee is}) = 0.8$$

A language model is a table

Prefix so far	coffee	tea	is	hot	cold	⟨end⟩	
	Next word						
nothing yet	0.6	0.4	0	0	0	0	
coffee	0	0	1.0	0	0	0	
tea	0	0	1.0	0	0	0	
coffee is	0	0	0	0.8	0.2	0	sums to 1
tea is	0	0	0	0.5	0.5	0	
coffee is hot	0	0	0	0	0	1.0	
⋮	⋮	⋮	⋮	⋮	⋮	⋮	

$$\boxed{\Pr(\,\text{hot} \mid \text{coffee is}\,) = 0.8}$$

Generating text is walking the table

Prefix so far	coffee	tea	is	hot	cold	⟨end⟩
nothing yet	0.6	0.4	0	0	0	0
coffee	0	0	1.0	0	0	0
tea	0	0	1.0	0	0	0
coffee is	0	0	0	0.8	0.2	0
tea is	0	0	0	0.5	0.5	0
coffee is hot	0	0	0	0	0	1.0
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Next word

▶ read this row

Text so far

Generating text is walking the table

Prefix so far	coffee	tea	is	hot	cold	⟨end⟩
nothing yet	1 0.6	0.4	0	0	0	0
coffee	0	0	1.0	0	0	0
tea	0	0	1.0	0	0	0
coffee is	0	0	0	0.8	0.2	0
tea is	0	0	0	0.5	0.5	0
coffee is hot	0	0	0	0	0	1.0
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Next word

▶ read this row

Text so far coffee

Generating text is walking the table

Prefix so far	coffee	tea	is	hot	cold	⟨end⟩
nothing yet	1 $\boxed{0.6}$	0.4	0	0	0	0
coffee	0	0	2 $\boxed{1.0}$	0	0	0
tea	0	0	1.0	0	0	0
coffee is	0	0	0	0.8	0.2	0
tea is	0	0	0	0.5	0.5	0
coffee is hot	0	0	0	0	0	1.0
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Next word (column heading over coffee … ⟨end⟩)

► read this row (beside "coffee is")

Text so far: coffee | is

Generating text is walking the table

Prefix so far	coffee	tea	is	hot	cold	⟨end⟩
nothing yet	1 0.6	0.4	0	0	0	0
coffee	0	0	2 1.0	0	0	0
tea	0	0	1.0	0	0	0
coffee is	0	0	0	3 0.8	0.2	0
tea is	0	0	0	0.5	0.5	0
coffee is hot	0	0	0	0	0	1.0
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Next word

▶ read this row

Text so far: coffee is hot

Generating text is walking the table

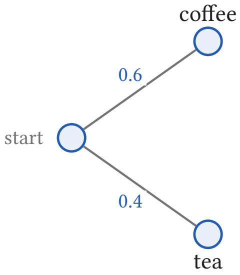
Prefix so far	coffee	tea	is	hot	cold	⟨end⟩
nothing yet	1 0.6	0.4	0	0	0	0
coffee	0	0	2 1.0	0	0	0
tea	0	0	1.0	0	0	0
coffee is	0	0	0	3 0.8	0.2	0
tea is	0	0	0	0.5	0.5	0
coffee is hot	0	0	0	0	0	4 1.0
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Next word

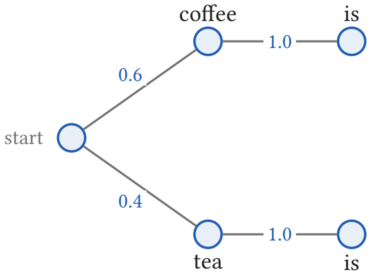
▶ read this row

Text so far coffee is hot and the model stops

One table, every text it can write



One table, every text it can write



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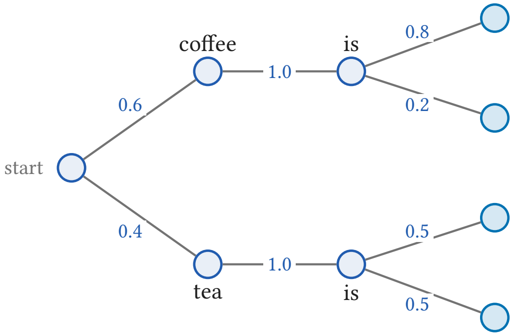

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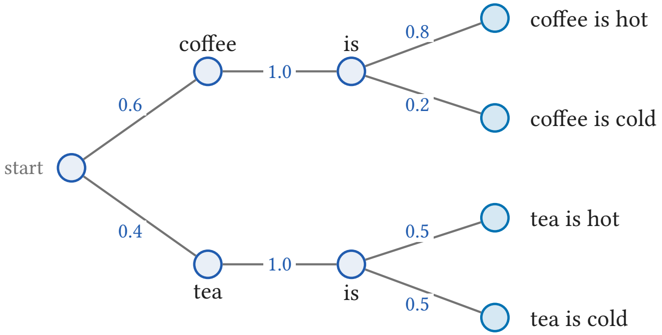

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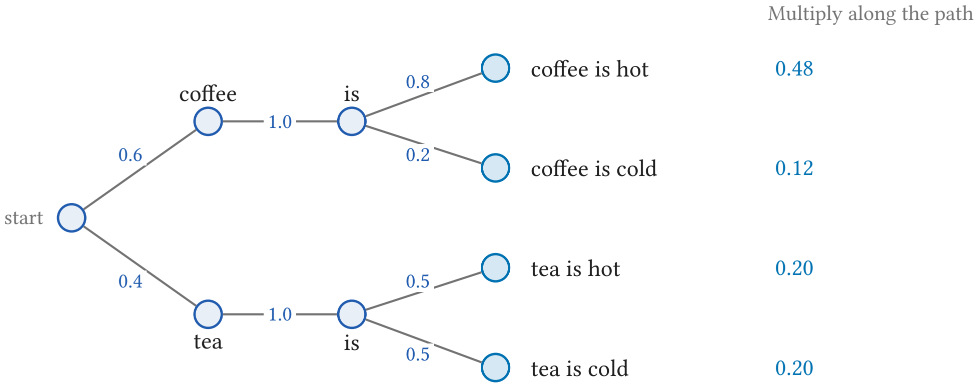

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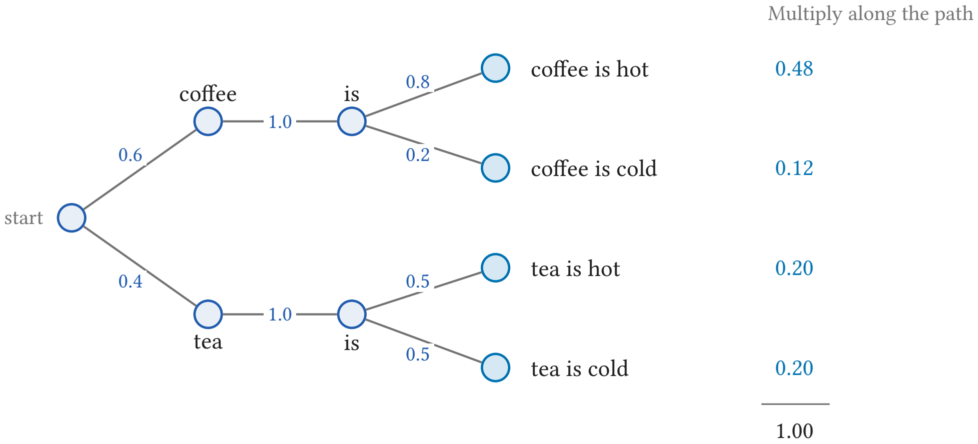


Section 3

Compressing the prefix

3

language model → LLM → reasoning model → tree of thought → agents

Real prefixes are long

Each report is a prefix, ending …so tomorrow will be. The next word finishes the sentence.

	sunny	rainy	snowy
It is cold and raining hard, …	0	0.3	0.7

Real prefixes are long

Each report is a prefix, ending …so tomorrow will be. The next word finishes the sentence.

	sunny	rainy	snowy
It is cold and raining hard, …	0	0.3	0.7
It is chilly and the air is dry, …	0.9	0	0.1

Real prefixes are long

Each report is a prefix, ending …so tomorrow will be. The next word finishes the sentence.

	sunny	rainy	snowy
It is cold and raining hard, …	0	0.3	0.7
It is chilly and the air is dry, …	0.9	0	0.1
It is warm and pouring with rain, …	0.2	0.8	0

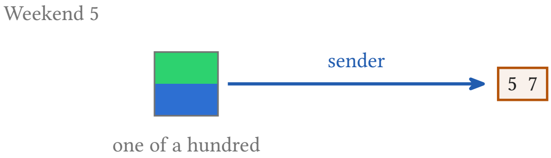
Real prefixes are long

Each report is a prefix, ending …so tomorrow will be. The next word finishes the sentence.

	sunny	rainy	snowy
It is cold and raining hard, …	0	0.3	0.7
It is chilly and the air is dry, …	0.9	0	0.1
It is warm and pouring with rain, …	0.2	0.8	0
Every other report anyone could write, in any wording, at any length, …	⋮	⋮	⋮

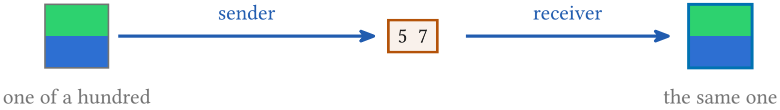
The toy table had six rows. This one has no last row.

The yogurt cup game, again



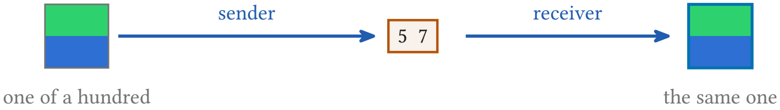
The yogurt cup game, again

Weekend 5



The yogurt cup game, again

Weekend 5

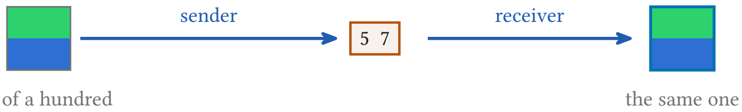


Today

It is cold and raining hard, ...

The yogurt cup game, again

Weekend 5

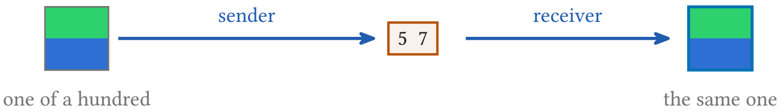


Today

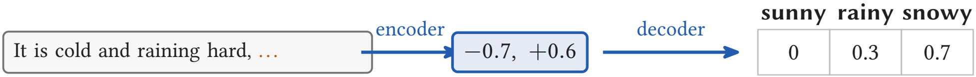
It is cold and raining hard, … $\xrightarrow{\text{encoder}}$ $\boxed{-0.7,\ +0.6}$

The yogurt cup game, again

Weekend 5

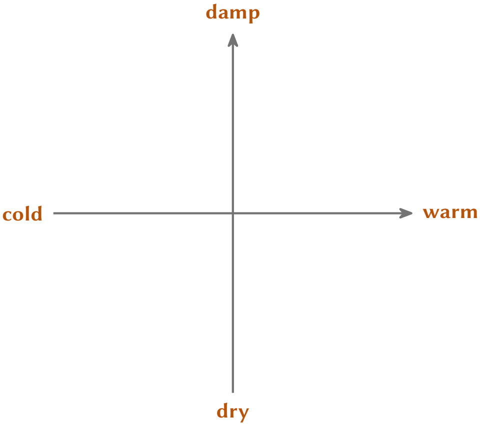


Today



One network, not two: the layers that read the prefix and the head that turns the result into a row, trained together by predicting the next word

The prefix becomes two numbers



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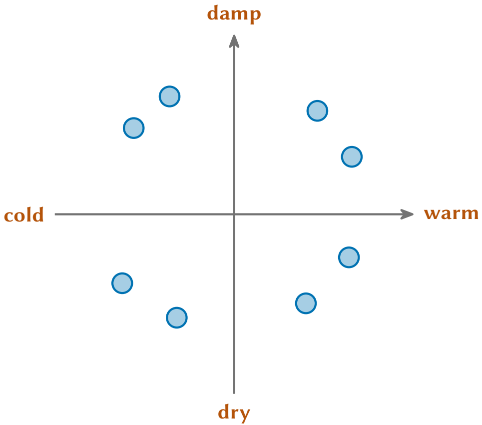

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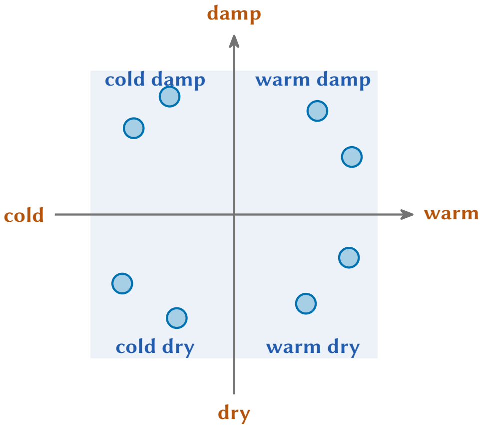
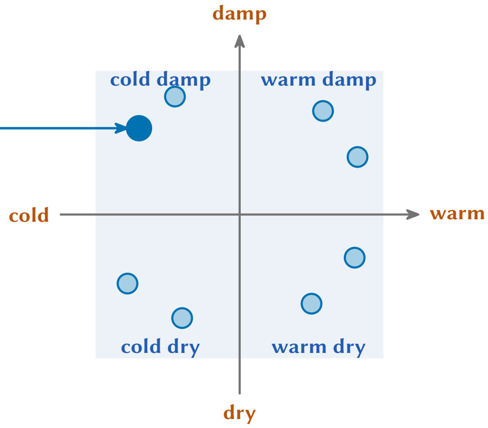


The prefix becomes two numbers

It is cold and raining hard, …



Every report ends the same way: …so tomorrow will be

A whole report in, two numbers out

Eight reports, four rows

Every report ends the same way: …so tomorrow will be

- It is cold and raining hard, …
- Freezing outside and the ground is wet, …
- It is chilly and the air is dry, …
- Icy morning and no rain at all, …
- It is warm and pouring with rain, …
- Hot outside and the ground is soaked, …
- It is hot and completely dry, …
- Mild morning and not a cloud anywhere, …

sunny	rainy	snowy

Eight reports, four rows

Every report ends the same way: …so tomorrow will be

- It is cold and raining hard, …
- Freezing outside and the ground is wet, …
- It is chilly and the air is dry, …
- Icy morning and no rain at all, …
- It is warm and pouring with rain, …
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- It is hot and completely dry, …
- Mild morning and not a cloud anywhere, …

sunny	rainy	snowy
0	0.3	0.7
0	0.3	0.7

cold damp

Eight reports, four rows

Every report ends the same way: …so tomorrow will be

- It is cold and raining hard, …
- Freezing outside and the ground is wet, …
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- Icy morning and no rain at all, …
- It is warm and pouring with rain, …
- Hot outside and the ground is soaked, …
- It is hot and completely dry, …
- Mild morning and not a cloud anywhere, …

sunny	rainy	snowy	
0	0.3	0.7	cold damp
0	0.3	0.7	
0.9	0	0.1	cold dry
0.9	0	0.1	

Eight reports, four rows

Every report ends the same way: …so tomorrow will be

- It is cold and raining hard, …
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- Mild morning and not a cloud anywhere, …

sunny	rainy	snowy	
0	0.3	0.7	cold damp
0	0.3	0.7	
0.9	0	0.1	cold dry
0.9	0	0.1	
0.2	0.8	0	warm damp
0.2	0.8	0	

Eight reports, four rows

Every report ends the same way: …so tomorrow will be

- It is cold and raining hard, …
- Freezing outside and the ground is wet, …
- It is chilly and the air is dry, …
- Icy morning and no rain at all, …
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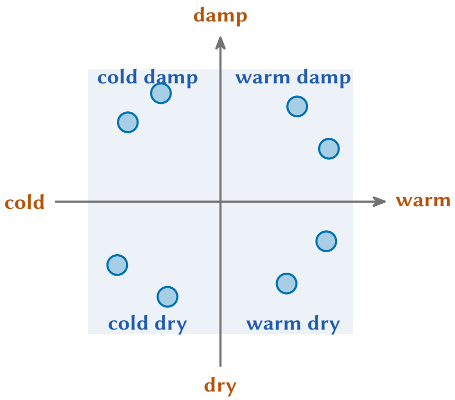
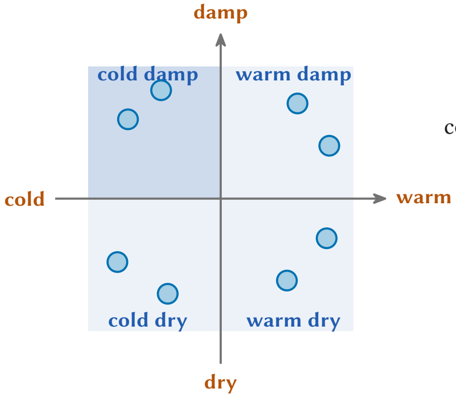
sunny	rainy	snowy	
0	0.3	0.7	cold damp
0	0.3	0.7	
0.9	0	0.1	cold dry
0.9	0	0.1	
0.2	0.8	0	warm damp
0.2	0.8	0	
0.9	0.1	0	warm dry
0.9	0.1	0	

One row for each region



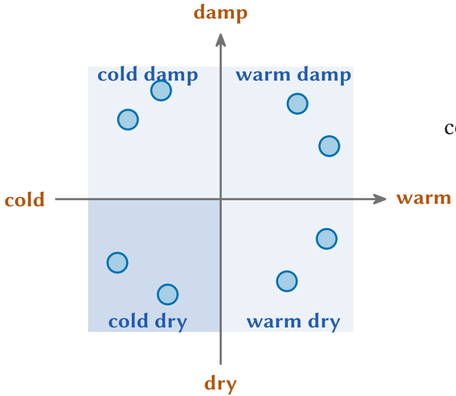
sunny	rainy	snowy

One row for each region



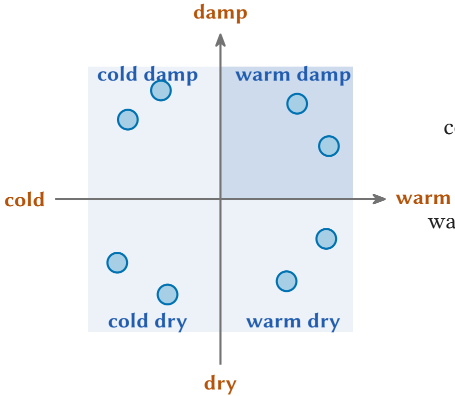
	sunny	rainy	snowy
cold damp	0	0.3	0.7

One row for each region



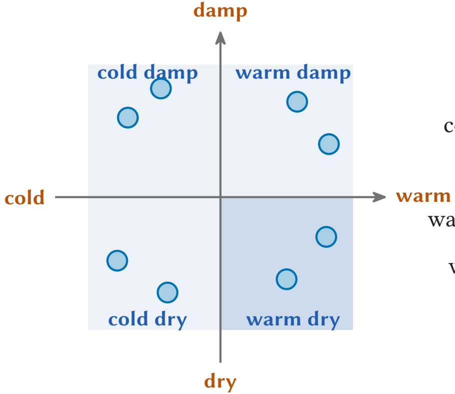
	sunny	rainy	snowy
cold damp	0	0.3	0.7
cold dry	0.9	0	0.1

One row for each region



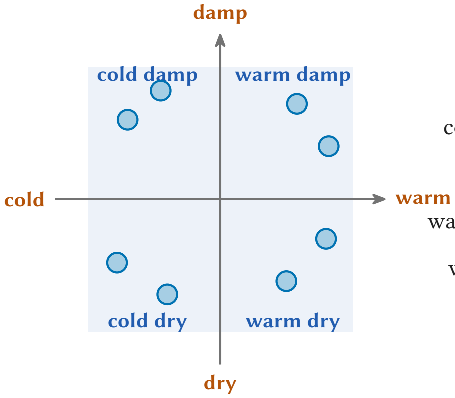
	sunny	rainy	snowy
cold damp	0	0.3	0.7
cold dry	0.9	0	0.1
warm damp	0.2	0.8	0

One row for each region



	sunny	rainy	snowy
cold damp	0	0.3	0.7
cold dry	0.9	0	0.1
warm damp	0.2	0.8	0
warm dry	0.9	0.1	0

One row for each region



	sunny	rainy	snowy
cold damp	0	0.3	0.7
cold dry	0.9	0	0.1
warm damp	0.2	0.8	0
warm dry	0.9	0.1	0

Cold alone gives 0.7 against 0.1. Damp alone gives 0.7 against 0.0. Only both give snowy.

Section 4

Why writing helps

4

language model > LLM > **reasoning model** > tree of thought > agents

The hop cipher

Start at cell 1. Write its letter. Move right by its number. Repeat.

A	R	O	G	K	E	L	I	N	S	T	M
3	5	1	2	4	3	2	6	2	3	4	2

The hop cipher

Start at cell 1. Write its letter. Move right by its number. Repeat.

A	R	O	G	K	E	L	I	N	S	T	M
3	5	1	2	4	3	2	6	2	3	4	2

straight off the strip A R O G K E L I N S T M → ?

The hop cipher

Start at cell 1. Write its letter. Move right by its number. Repeat.

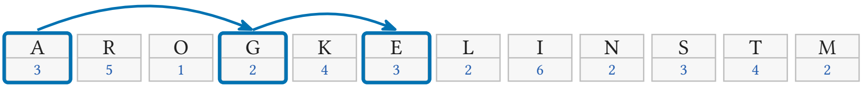
A	R	O	G	K	E	L	I	N	S	T	M
3	5	1	2	4	3	2	6	2	3	4	2

straight off the strip A R O G K E L I N S T M → ?

the recipe spells A

The hop cipher

Start at cell 1. Write its letter. Move right by its number. Repeat.

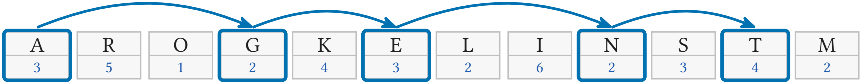


straight off the strip A R O G K E L I N S T M → ?

the recipe spells A G E

The hop cipher

Start at cell 1. Write its letter. Move right by its number. Repeat.



straight off the strip A R O G K E L I N S T M → ?

the recipe spells A G E N T

What the extra words buy

- Every hop is easy. There are simply too many in a row for one pass.

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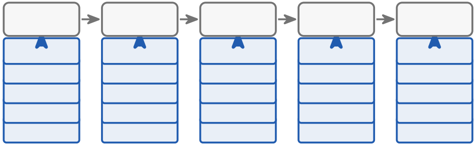

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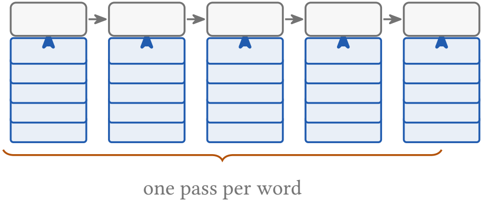

What the extra words buy

- Every hop is easy. There are simply too many in a row for one pass.
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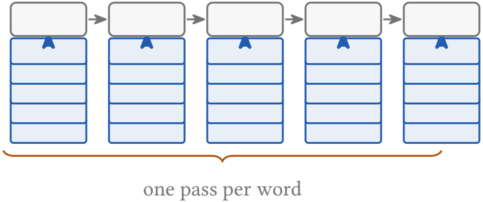
- The depth is fixed. The number of passes is not.
- With no intermediate words a transformer is confined to one shallow circuit class.



Merrill and Sabharwal. The Expressive Power of Transformers with Chain of Thought. ICLR 2024.
Feng, Zhang, Gu et al. Towards Revealing the Mystery behind Chain of Thought: A Theoretical Perspective. NeurIPS 2023.

Section 5

Training the model to reason

5

language model ➤ LLM ➤ **reasoning model** ➤ tree of thought ➤ agents

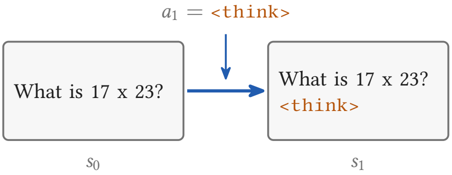
The chain of thought as a decision problem

What is 17 x 23?

s_0

state the prompt plus everything written so far, so the state grows by one token at every step

The chain of thought as a decision problem



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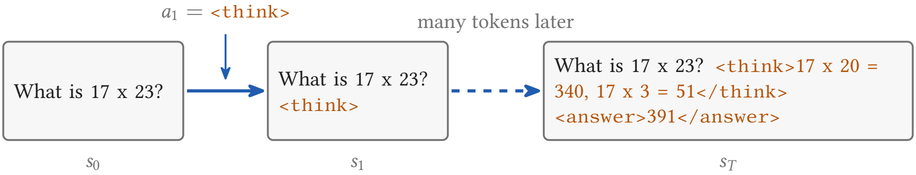
The chain of thought as a decision problem



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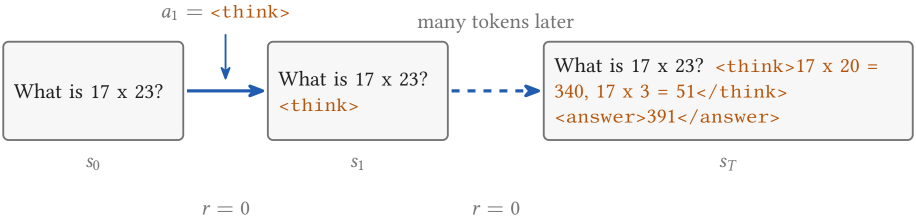
The chain of thought as a decision problem



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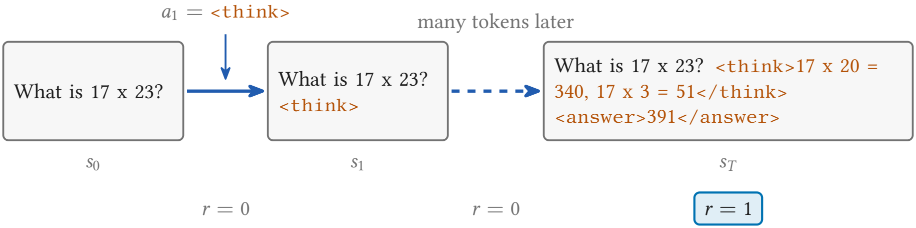

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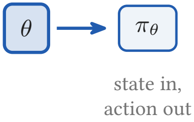
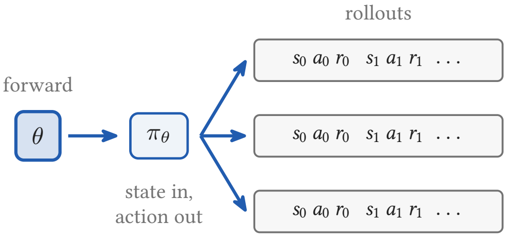
Guo, Yang, Zhang et al. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. Nature 645, 633 to 638, 2025.

How reinforcement learning trains a policy



How reinforcement learning trains a policy



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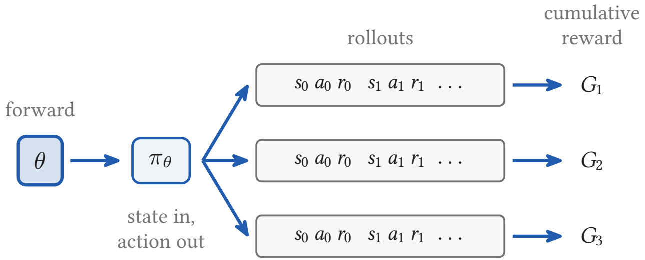
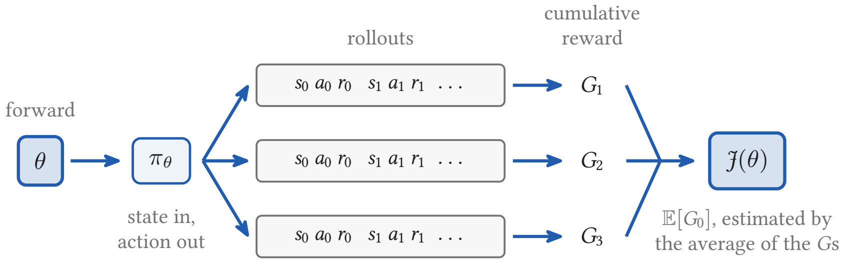


How reinforcement learning trains a policy



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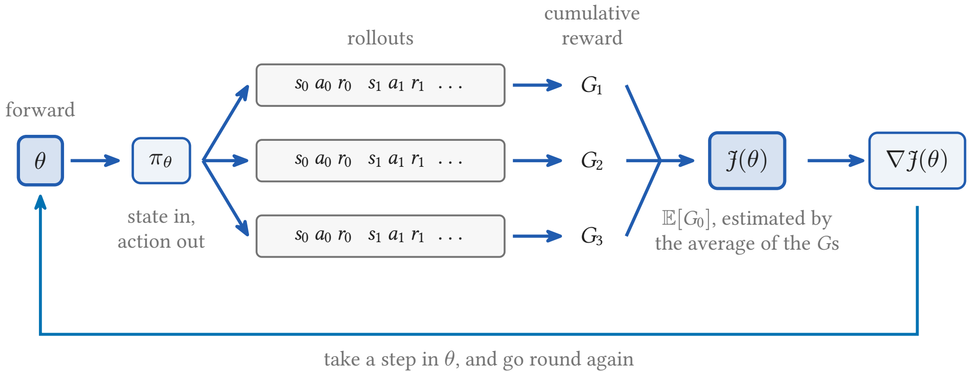

Weekend 4, the same loop with a point in place of a policy: the grasshopper on a hill it cannot see

The gradient, term by term

$$\nabla_\theta \mathcal{J}(\theta) = \mathbb{E}\Big[\sum_{t=0}^{T-1} \gamma^{t} \, \Psi_t \; \nabla_\theta \log \pi_\theta(a_t \mid s_t) \Big]$$

The expectation is over episodes generated by π_θ; in practice, the average over the rollouts just collected.

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the score
which way to nudge θ so that a_t becomes more likely at s_t

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the multiplier
how far to push,
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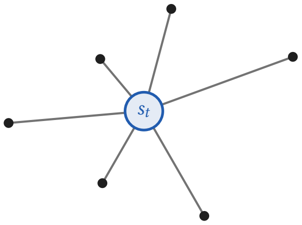
The logarithm makes the nudge relative: a rare action that pays off moves further than one the policy already takes.

How much to believe the direction

- One spoke per action available at s_t, as long as that action turned out to be worth.



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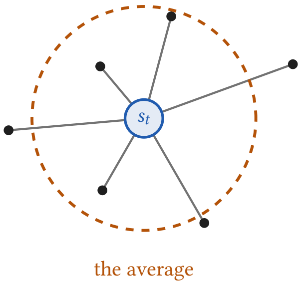

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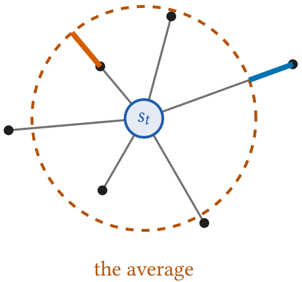

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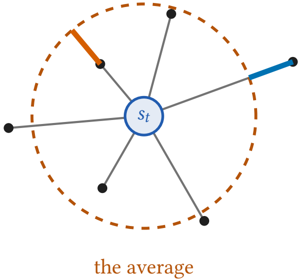

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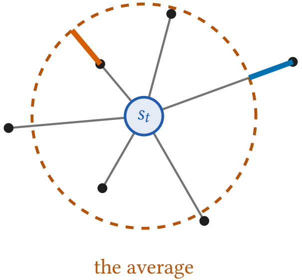

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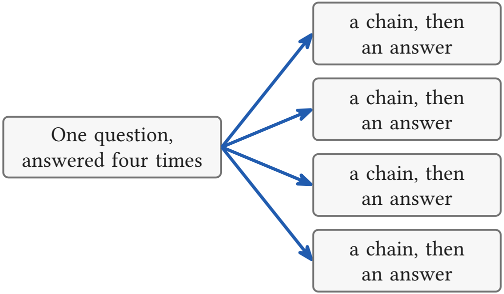
For a model that reasons, the **actor** is the language model itself. And the critic?

The group is the baseline



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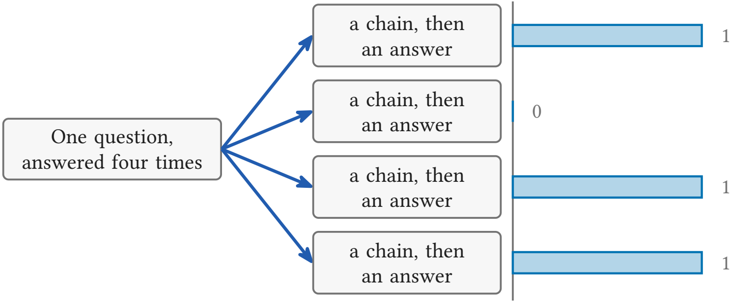

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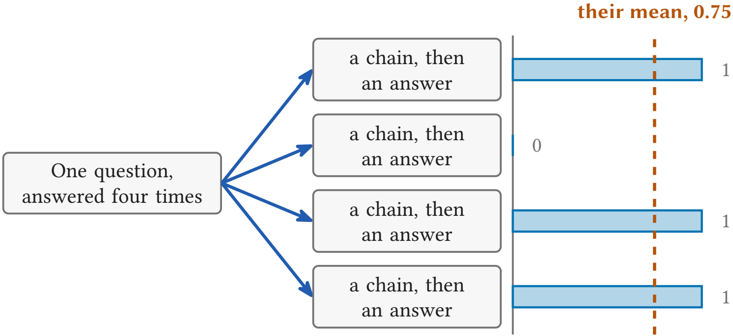

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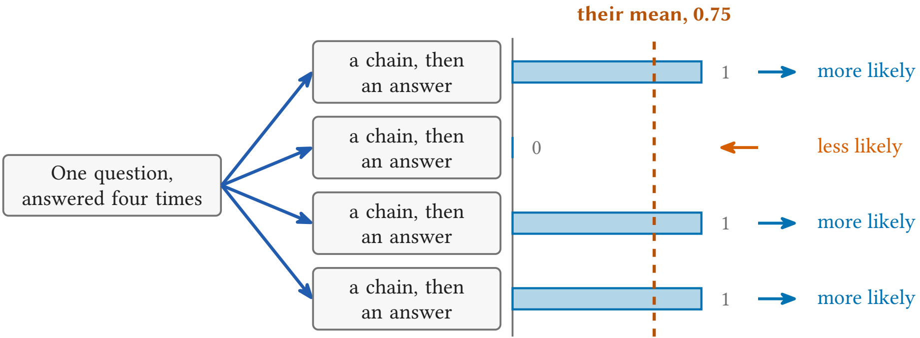


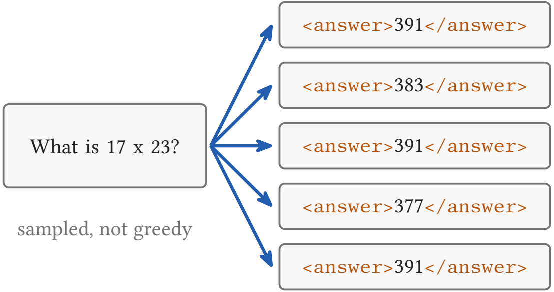
Guo, Yang, Zhang et al. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. Nature 645, 633 to 638, 2025.

Asking the same question several times

What is 17 x 23?

sampled, not greedy

Asking the same question several times



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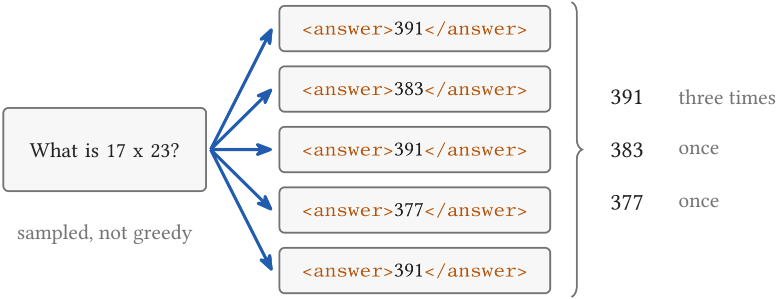

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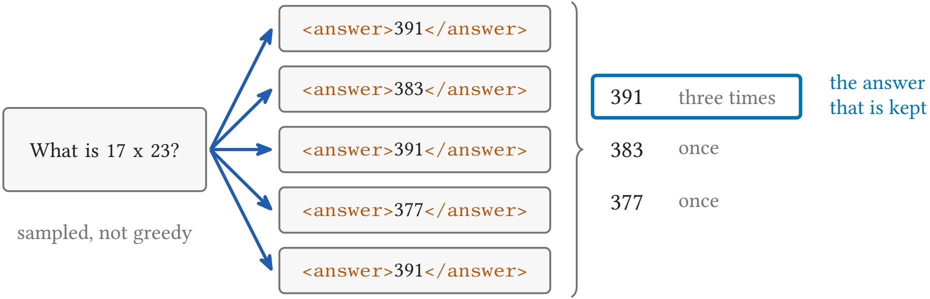

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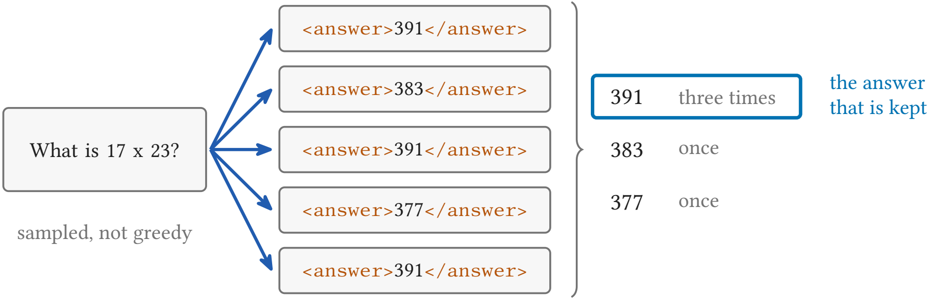

- No weight changes. This is a way of decoding, not a way of training.
- One vote costs five whole answers, so voting and not voting are not the same measurement.

Wang, Wei, Schuurmans et al. Self-Consistency Improves Chain of Thought Reasoning in Language Models. ICLR 2023.

Reasoning from pure reinforcement learning

- DeepSeek-R1-Zero starts from the base model: no supervised fine-tuning, no worked examples of reasoning.

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AIME 2024, pass@1

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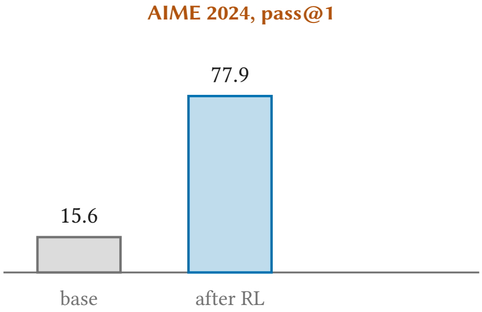
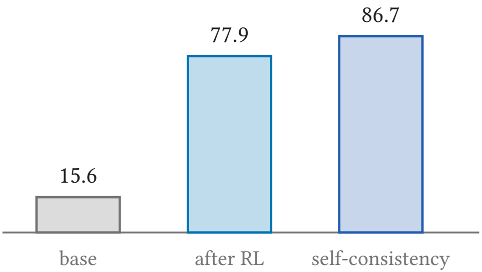


Reasoning from pure reinforcement learning

- DeepSeek-R1-Zero starts from the base model: no supervised fine-tuning, no worked examples of reasoning.
- Longer chains and self-checking appeared on their own.

AIME 2024, pass@1



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Section 6

From prompting to agents

6

language model → LLM → reasoning model → tree of thought → agents

From a table to an agent

agents	LATS	all of them, inside a search
	Monte Carlo Tree Search	look ahead before moving
	ReAct	think, act, observe
prompting	Tree-of-Thought	branch, score, keep the best
	Chain-of-Thought	write the reasoning first
the model	Reasoning model	trained by RL to reason before answering
	Large language model	a transformer compresses any prefix
	Language model	predict the next word

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one wrong step poisons the path

From a table to an agent

Group	Layer	Description	Remark
agents	LATS	all of them, inside a search	at 10:00
	Monte Carlo Tree Search	look ahead before moving	**next, this morning**
	ReAct	think, act, observe	
prompting	Tree-of-Thought	branch, score, keep the best	
	Chain-of-Thought	write the reasoning first	one wrong step poisons the path
the model	Reasoning model	trained by RL to reason before answering	
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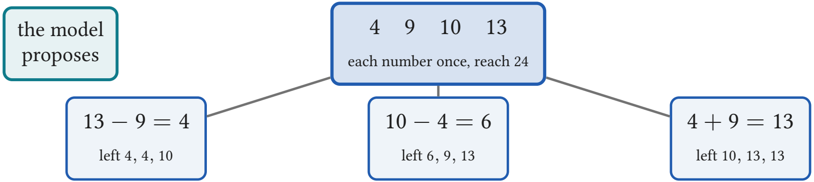
Wei et al. 2022 (Chain-of-Thought), Yao et al. 2023 (Tree of Thoughts, ReAct), Guo et al. 2025 (DeepSeek-R1), Zhou et al. 2024 (LATS).

Tree-of-Thought: one problem, worked

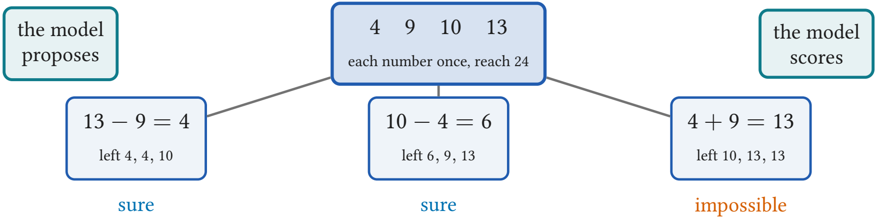
4 9 10 13

each number once, reach 24

Tree-of-Thought: one problem, worked



Tree-of-Thought: one problem, worked



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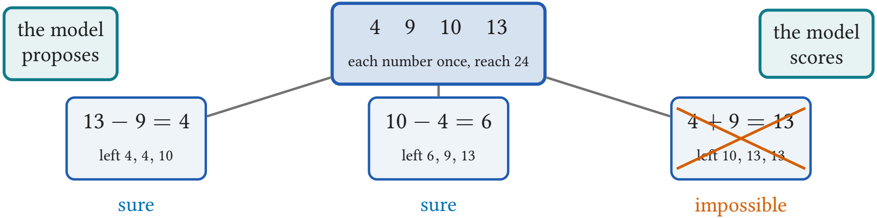

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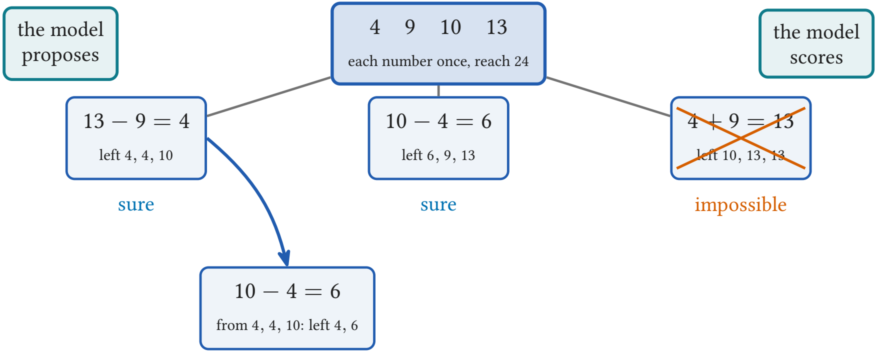

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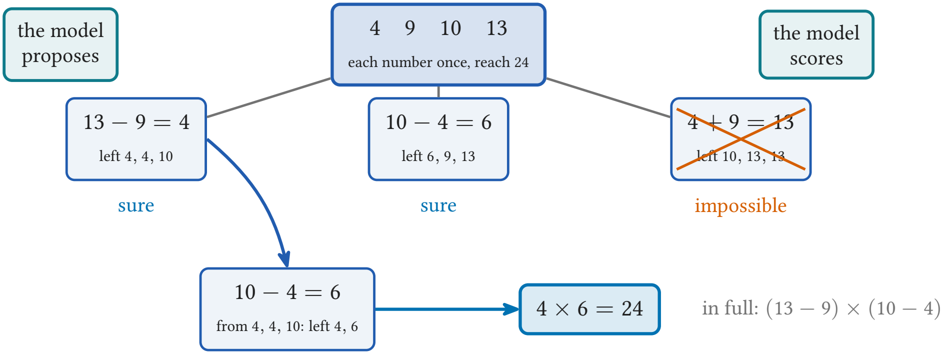

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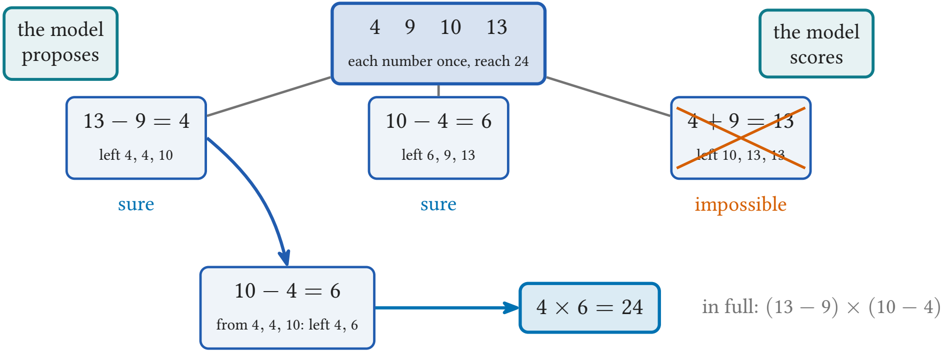
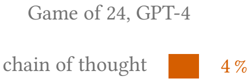


three steps deep, five candidates kept at each one

Tree-of-Thought: what the search buys



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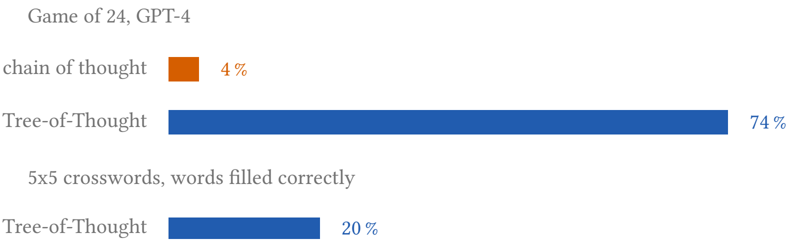

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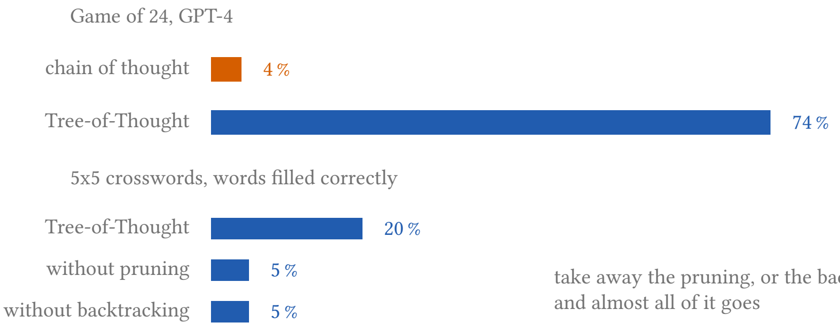
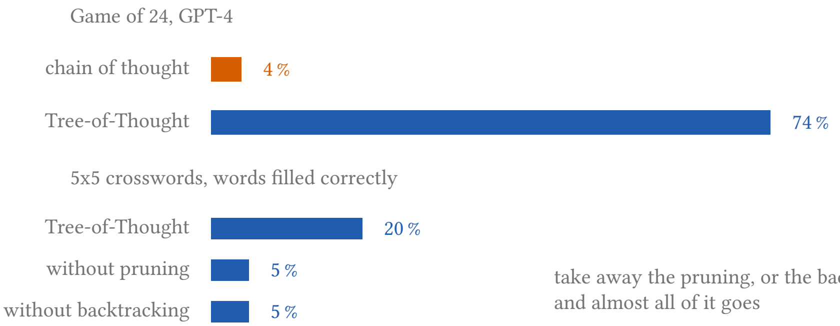


take away the pruning, or the backtracking, and almost all of it goes

Tree-of-Thought: what the search buys



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The authors search the tree breadth first and depth first, and call those relatively simple. They leave A* and Monte Carlo Tree Search to future work. That is the rest of this morning.

Backup

frames kept for questions

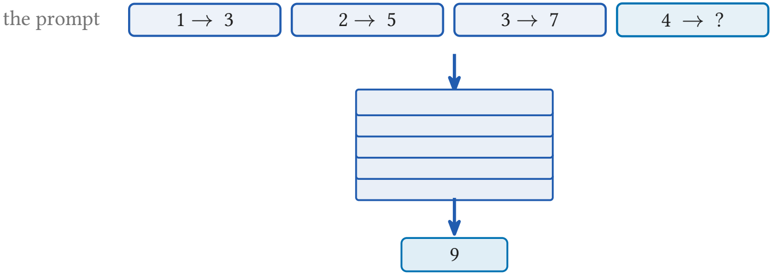
A regression problem in the prompt

the prompt	$1 \rightarrow 3$	$2 \rightarrow 5$	$3 \rightarrow 7$

A regression problem in the prompt

the prompt | $1 \to 3$ | $2 \to 5$ | $3 \to 7$ | $4 \to ?$

A regression problem in the prompt



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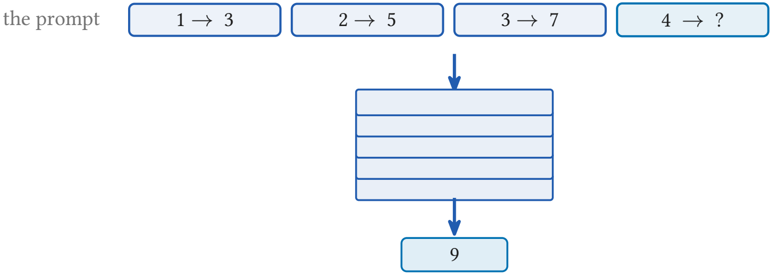
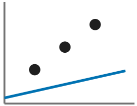


One pass. No weight changed, and the pairs exist only in the prompt

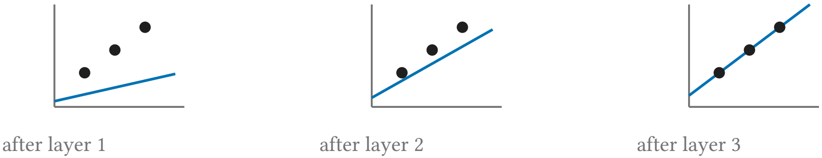
The layers can do the fitting



after layer 1

- By explicit construction, the layers can carry out gradient descent.

The layers can do the fitting



- By explicit construction, the layers can carry out gradient descent.
- So one forward pass can solve a basic regression problem, shown for linear models.

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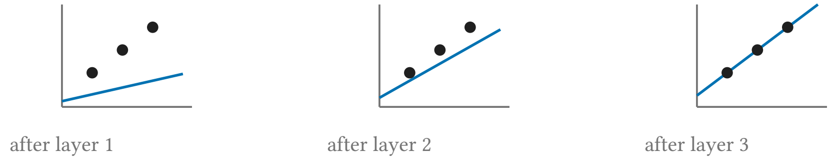

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Akyurek, Schuurmans, Andreas et al. What learning algorithm is in-context learning? Investigations with linear models. ICLR 2023.
von Oswald, Niklasson, Randazzo et al. Transformers Learn In-Context by Gradient Descent. ICML 2023.